Solution Requirements

Date	15 July 2026
Team ID	SWTID-2026-2378
Project Name	Credit card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	**Performance & Speed**	The system should generate credit card	Prediction result displayed within **2–3 seconds**.

- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Allow users to enter applicant details through a web form.	High
2	Input Validation	Validate all mandatory fields and data formats before submission.	High
3	Data Processing	Convert user input into the format required by the ML model.	High
4	Prediction	Predict whether the credit card application is Approved or Rejected using the Random Forest model.	High
5	Result Display	Display the prediction result to the user in a user-friendly format.	High
6	Model Loading	Load the trained machine learning model during application startup.	High
7	Error Handling	Display appropriate error messages for invalid inputs or system errors.	Medium
8	User Interface	Provide an easy-to-use and responsive web interface.	Medium

Step 2: Non-Functional Requirements (NFR)

		approval predictions quickly after the user submits the application.	
2	**Scalability**	The system should support multiple users accessing the application simultaneously without significant performance degradation.	Supports **100+ concurrent users** with stable performance.
3	**Security & Data Privacy**	The system should protect applicant information and validate all user inputs to prevent unauthorized access or invalid data.	HTTPS deployment, server-side input validation, and secure storage of the ML model.

4	**Reliability & Availability**	The application should provide consistent predictions and remain available during normal operation.	**99% uptime** with consistent prediction results for identical inputs.
5	**Usability & Accessibility**	The application should have a simple, responsive, and easy-to-use interface for all users.	User can complete an application in **under 5 minutes**; compatible with desktop and mobile browsers.
6	**Maintainability & Portability**	The system should allow easy model updates and deployment on different platforms without major code changes.	Model can be replaced by updating the `.pkl` file; deployable on **Render**, **Railway**, or any Flask-supported cloud platform.